Dual-VT Based Content Addressable Memory Design for Low Power Applications

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***Abstract*—Content Addressable Memory (CAM) is a high-speed hardware structure that performs fully parallel search by comparing an input data against all the stored data simultaneously. The parallel activation of transistors during each search cycle results in significant dynamic power dissipation, but there is large static power dissipation that arises from the leakage current from each transistors when it is even in off state. This paper presents a low-power 8×8 NOR-type CAM array designed in UMC 65 nm CMOS technology by integrating a dual threshold-voltage (Dual-VT) strategy with the Hybrid Self-Controlled Precharge-Free (HSCPF) architecture [4]. The cross-coupled inverters of the 6T SRAM are implemented with High-VT (HVT) transistors to suppress static leakage power, while the access and search transistors use Low-VT (LVT) devices to preserve fast write and search performance. Transient simulations in Cadence Virtuoso at VDD = 1.2 V demonstrate an average search power of 0.100 mW , a search delay of 48.8 ps, and a Power-Delay Product (PDP) of 0.07623 fJ/bit/search in the match case scenario. The proposed design achieves 53.05 % power reduction, 34.05 % delay reduction, and 69.98 % PDP improvement compared with the HSCPF CAM architecture, validating the effectiveness of transistor-level Dual-VT partitioning for energy-constrained CAM applications.**

***Index Terms*—Content Addressable Memory, Dual Threshold Voltage, Low Power VLSI, HSCPF Architecture, NOR-type Match Line, Leakage Reduction, 6T SRAM, UMC 65 nm CMOS**

I. INTRODUCTION

Content Addressable Memory is a high-speed parallel search hardware that performs comparison between the input search data and the stored data and provides the address corresponding to the matched data [2]. Due to its high-speed search capability CAM is widely used in network routers, packet classification, image processing and many more. This high throughput parallel search mechanism comes with high power consumption due to the presence of large number of search line and match line circuitry that switch simultaneously during each search operation.

Conventional CAM architectures are broadly classified into NOR-type and NAND-type match line structures [2]. NOR-type CAM offers high search speed due to its short discharge path but suffers from large dynamic power consumption because most match lines discharge during evaluation. On the other hand, NAND-type CAM reduces dynamic power but experiences increased search delay and charge-sharing issues due to its series-connected structure. To overcome these limitations, several precharge-free [1] and self-controlled architectures have been proposed. Among them, Hybrid Self-controlled Precharge Free CAM architecture is the latest architecture with less power dissipation, improved performance and reduced area overhead. Although this architecture focused on reducing the dynamic power consumption the leakage power remains a significant concern as we scale down the technology nodes.

Power dissipation in a CAM array has two principal components.*Dynamic power* arises from the charging and discharging of the match lines, search lines, and internal bit-line nodes during each search cycle. *Static power* mainly the leakage power that originates mainly from the cross-coupled inverter pair of the 6T SRAM storage cell, which remains permanently biased to retain stored data. Due to its always on state leakage power dissipation dominates causing the overall CAM power consuming. [2]. Several architectural innovations have been proposed to mitigate dynamic power in CAM. Precharge-Free (PF) CAM [1] eliminates the explicit match-line precharge transistor, reducing the capacitive switching energy. The Self-Controlled Precharge-Free (SCPF) architecture [3] adds a self-timed control circuit to further suppress unnecessary match-line transitions. The Hybrid Self-Controlled Precharge-Free (HSCPF) design [4] combines hybrid control logic with precharge free circuits to achieve good Power-Delay Product. However,these works treat all transistors within the CAM cell at a single threshold voltage and therefore do not specifically address the leakage component arising from the continuously-on cross coupled inverters.

Multi-threshold voltage design is a well-established technique in standard-cell synthesis, where a cell library offers transistor of two or more threshold voltages. High-VT (HVT) cells reduce leakage at the cost of speed; Low-VT (LVT) cells provide speed at the cost of leakage. The central insight of the present work is that within a single CAM cell these two requirements that is the cross coupled inverter is a leakage critical block since its always ON while the access transistors and the search transistors are performance critical blocks. This observation motivates a within-cell Dual-VT assignment that applies HVT devices selectively to the latch inverters and LVT devices to the access and search transistors.

II. Design Methodology

A. Conventional CAM Architecture

Content Addressable Memory consists of storage element that is mainly the 6T-SRAM cell consisting of Two cross coupled inverters that stores data and the tow access transistors that forms the integral part of read and write operation. For the search circuitry in the CAM we have the two more transistors that through which data is searched within the SRAM.There is also a match-line circuitry that can be of two types NAND or NOR type match-lines. Here in this work NOR type match-line is used, when the searched data is present in the storage the match-line will be high and vice versa. These match-lines should be precharged high in-order to facilitate correct sensing of the match-line voltage. This precharging of Match-lines before every search operations contributes to major source of dynamic power.

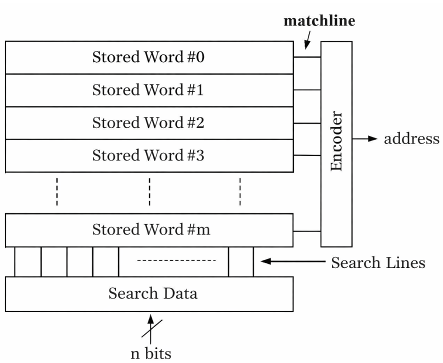

Fig. 1. Conventional Content Addressable Memory (CAM) architecture

In [1], proposed an 8T-CAM architecture which removes the precharging of Match-lines of the CAM cell therby reducing the dynamic power dissipation of the circuit. Which is further modified in [4] which is HSCPF CAM architecture.

B. NOR-Type Matchline

In a conventional Content Addressable Memory (CAM), the NOR-type matchline architecture is widely adopted due to its simplicity and fast search operation. In this scheme, each CAM cell contributes a parallel comparison path between the stored data and the search input. The matchline (ML) is typically precharged to logic high (VDD) during the precharge phase. During the evaluation phase, depending on the comparison result, the matchline either remains high (indicating a match) or discharges to ground (indicating a mismatch).

In the NOR-type configuration, mismatch detection is dominant, meaning that even a single bit mismatch can discharge the matchline through the evaluation NMOS transistor network. Each bit cell includes comparison transistors controlled by the stored data and search lines (SL and SL). If the stored bit differs from the search input, a conducting path is created between the matchline and ground, resulting in a rapid discharge. Conversely, when all bits match, no discharge path is activated, and the matchline remains at its precharged high level.

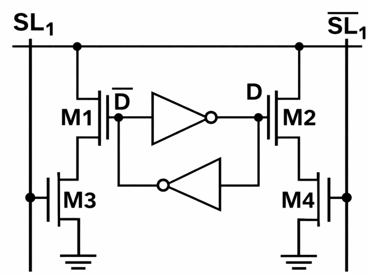

Fig. 2. NOR Type Match-Line

The key advantage of the NOR-type matchline lies in its low evaluation delay for mismatch cases, as the discharge path is directly connected to ground through parallel transistors. However, this architecture suffers from high dynamic power consumption, primarily due to frequent precharging of matchlines and unnecessary switching activity during search operations. Additionally, the large capacitive load of long matchlines can increase delay and power dissipation.

C. 8T - Precharge Free CAM

An 8-transistor (8T) Content Addressable Memory (CAM) cell is an enhanced memory structure designed to perform high-speed parallel search operations while improving power efficiency compared to conventional CAM designs. The 8T CAM cell typically consists of a 6T SRAM storage core combined with two additional transistors that enable comparison between stored data and search inputs.The 6T SRAM cell forms the storage element, comprising two cross-coupled inverters and access transistors that store the bit and its complement (D and $\overline{D}$). The additional two transistors are used to implement the search and match functionality, connecting the stored data to the matchline (ML) through the search lines (SL and $\overline{SL}$).

During the search operation, the search data are applied to the search lines. The operation of the 8T CAM cell can be described as,If the stored bit matches the search bit, the

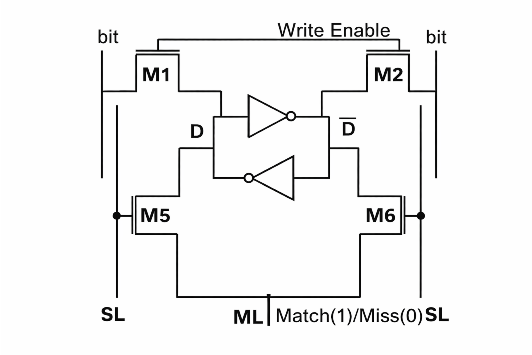

Fig. 3. 8T Precharge Free CAM

comparison transistors do not create a discharge path, allowing the matchline to remain charged (logic HIGH).If there is a mismatch, a conduction path is formed between the matchline and ground, causing the matchline to discharge (logic LOW). Thus, the matchline voltage indicates whether the stored word matches the search word.

D. Hybrid Self-Controlled Precharge Free (HSCPF) CAM

HSCPF CAM employs a hybrid charge control mechanism, where the matchline evaluation is controlled by the charge states of two consecutive CAM cells. Each HSCPF unit consists of two 6T SRAM storage cells along with comparison transistors and a shared charge control circuit. The internal nodes of these two cells, denoted as SN-1 and SN,are used to control the matchline through a pair of transistors. During

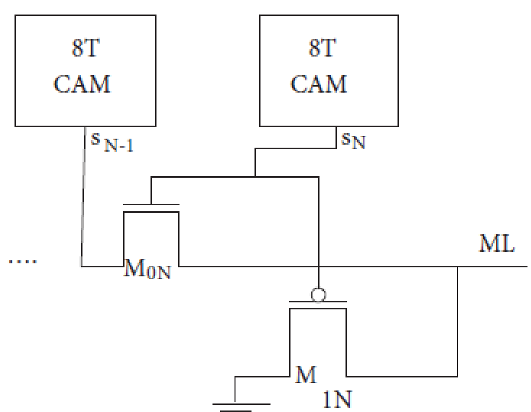

Fig. 4. Hybrid Self-Controlled Pre-charge Free CAM (HSCPF)

the search operation, the matchline behavior depends on the combined charge values of nodes S0 and S1.If both nodes are at logic high, indicating that the stored bits match the search input, the charge propagates through the control circuitry and the matchline is driven to logic high, representing a match condition. In contrast, if any one of the bits mismatches, at least one of the nodes becomes low, thereby preventing charge propagation and causing the matchline to evaluate to logic low, indicating a mismatch. This grouped evaluation reduces unnecessary switching activity and improves efficiency.

III. Proposed Dual-VT CAM Design

A. Dual-VT Assignment strategy

The fundamental unit of the proposed array is an 8-transistor (8T) NOR-type CAM cell, as illustrated conceptually in Fig. 3. The overall design workings are closely aligned with the HSCPF CAM architecture. The cell comprises:

- A standard 6T SRAM cell consisting of two cross-coupled CMOS inverters (4 transistors: MP1, MP2, MN1, MN2) and two NMOS access transistors (M1, M2) controlled by the Word Line (WL).
- Two NMOS search transistors (M5, M6) connected in series between the Match Line (ML) and ground, controlled by the Search Lines (SL and $\overline{\text{SL}}$) and the storage nodes D and $\overline{\text{D}}$.

During a search operation, the SL and $\overline{\text{SL}}$ lines are driven with the query bit and its complement respectively. A mismatch condition (stored bit not equal to search bit) turns on one search transistor pair, discharging the ML toward ground. A match condition keeps both search paths off, leaving ML at its high initial state. In a NOR-type array, ML is high if and only if all bits in the row match the query a single mismatch anywhere in the row pulls ML low.

The proposed design introduces a systematic threshold-voltage assignment within the 8T cell implemented in the UMC 65 nm CMOS process, which provides both HVT and LVT transistor within the same technology node. The assignment is summarized in Table I.

TABLE I
Dual-VT Transistor Assignment in the Proposed 8T CAM Cell

Transistor	Role	VT	Reason
MP1, MP2	PMOS Latch	HVT	Leakage suppression in SRAM cell
MN1, MN2	NMOS Latch	HVT	Leakage suppression in SRAM cell
M1, M2	Access (WL)	LVT	Fast write access
M5, M6	Search (SL)	LVT	Fast ML discharge and speed-critical

1) HVT Cells for the cross coupled inverter latch: The cross-coupled inverters that form the SRAM latch are permanently powered to retain the stored data. They draw subthreshold leakage current $I_{\text{leak}} \propto e^{-V_T/(nV_{th})}$ continuously, regardless of search or write activity. The assignment of HVT cells to these four transistors increases V_T and is exponentially suppressed I_{leak} and therefore reduce the static power dissipation. Since the cross coupled inverters operate at DC during data storage and their switching speed is not on the critical path of any search or write operation,the slow activity of the cross coupled due to HVT cells are acceptable.

2) LVT for Access Transistors (M1, M2): The access transistors connect the bit lines (BL, $\overline{\text{BL}}$) to the storage nodes during write operations. When a new data bit is written into the cell, the access transistors must supply sufficient current to flip the state of the cross-coupled inverters, which actively resist any change in their stored value. If the drive current is too weak, the inverter latch retains its old value and the write operation fails. By assigning LVT devices to the access transistors, a higher drive current is achieved at the same gate voltage, ensuring that the latch state is successfully overwritten within the write pulse duration at a supply voltage of 1.2 V.

3) LVT for Search Transistors (M5, M6): During a search operation, when the stored bit does not match the search input, the search transistors are responsible for discharging the match line (ML) from its high state to ground, which signals a mismatch condition. The speed at which this discharge happens directly determines the search delay of the CAM. If the discharge is slow, the match line takes longer to reach a detectable low voltage level, increasing the overall search latency. By assigning LVT devices to the search transistors, a higher drain current (I_{DS}) is available for the same applied gate voltage, which allows the match line capacitance to discharge more quickly. This faster discharge reduces the time taken to detect a mismatch, directly contributing to the improved search delay of 48.8 ps reported in Section IV.

B. HSCPF Architecture Integration

The dual-VT 8T cells used for the design of the HSCPF architecture [4]. In the HSCPF framework, NOR-type match lines operate without a dedicated precharge transistor; instead, a hybrid self-controlled circuit monitors the search lines and conditionally enables match-line evaluation. This eliminates the power-hungry precharge phase while maintaining robust match detection through the hybrid control logic.

CAM power dissipation has two principal components, dynamic power which arises from the switching activity of the match lines and search lines during each search operation, and static power, which originates from the sub-threshold leakage current of the transistors even when the circuit is idle. Conventional CAM architectures typically address only one of these components at a time. The proposed design tackles both simultaneously by combining two complementary techniques. The HSCPF architecture eliminates the dedicated match line precharge phase, which is the dominant source of dynamic power in conventional CAM designs. The Dual-VT cell assignment selectively places HVT transistors in the cross-coupled inverters of the SRAM latch, where leakage is the primary concern, suppressing the static power dissipation. Together, these two techniques complement each other and form a unified design strategy that reduces both dynamic and static power without compromising search speed.

C. 8×8 CAM Array Implementation

An 8×8 CAM array was designed in Cadence Virtuoso using the UMC 65 nm pdk at VDD = 1.2 V. The array comprises 64 dual-VT 8T cells arranged in 8 rows and 8 columns with shared word lines, bit lines, search lines one match line per row (ML[0:7]), they are designed following the HSCPF architecture.

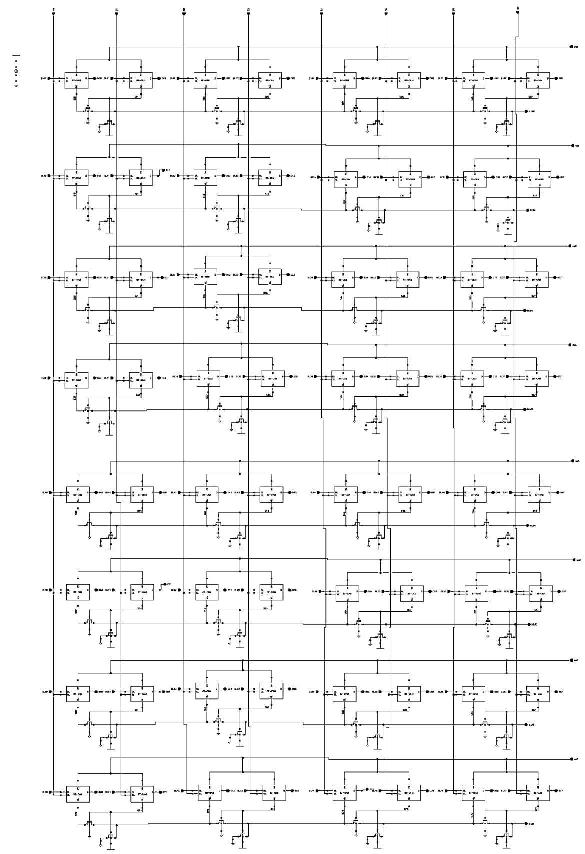

Fig. 5. Proposed 8×8 Dual-VT CAM Array schematic (Cadence Virtuoso, UMC 65 nm)

IV. Simulation Results

A. Simulation Setup

All simulations were performed using Cadence Virtuoso Spectre transient analysis at VDD = 1.2 V. The following 8-bit test patterns were written sequentially into the eight rows of the CAM array:

Word 0: `1111 1111`	Word 4: `1011 1100`
Word 1: `1000 1100`	Word 5: `1110 0101`
Word 2: `1100 1100`	Word 6: `0000 0000`
Word 3: `0000 0011`	Word 7: `0010 0100`

Two search scenarios were evaluated: a *match case* (search data `1111 1111`, present in Word 0) and a *mismatch case* (search data `1010 1101`, absent from all rows). The mismatch case represents the worst-case power and delay scenario because all match lines discharge.

B. Match Case Results

When the search word `1111 1111` is applied, all eight search lines are simultaneously asserted to 1.0 V. Since Word 0 stored in the first row exactly matches the searched data, ML[0] rises to approximately 900 mV, confirming a match, as shown in Fig. 6. All remaining match lines ML[1]–ML[7] are pulled to near-ground levels, correctly indicating mismatch for the other rows.

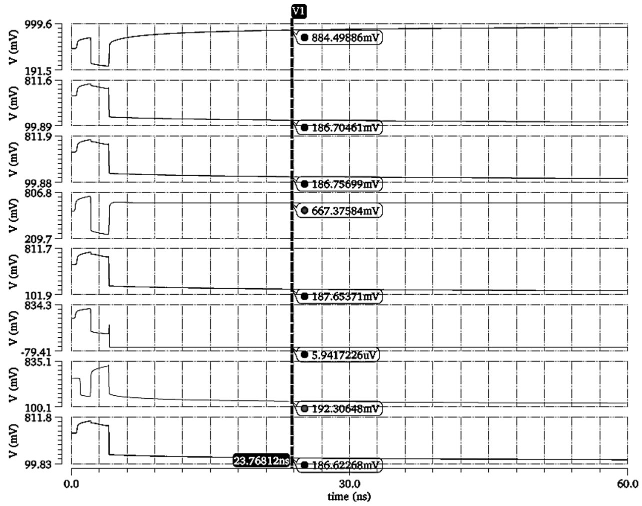

Fig. 6. Match case ML waveforms

The total power during the match operation is shown in Fig. 7, with sharp transient spikes at the moment of search line switching, settling to a low steady state after the match lines reach their final values. The average power consumption during the match case is 0.100,mW. The search delay is calculated which is 48.8,ps.

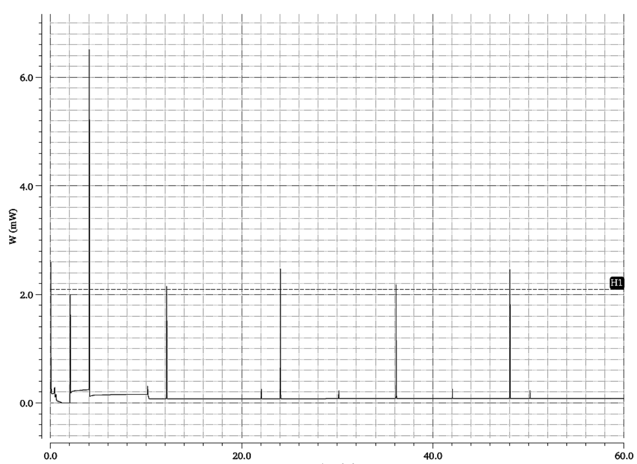

Fig. 7. Power plot for the match case

C. Mismatch Case Results

When search bit `1010 1101` is applied, all eight match lines discharge to near ground levels (below 10 μV for fully-mismatching rows), indicating that the searched word is not present in the memory. The search delay measured is 140.24 ps. The average power during the mismatch operation 0.164 mW.

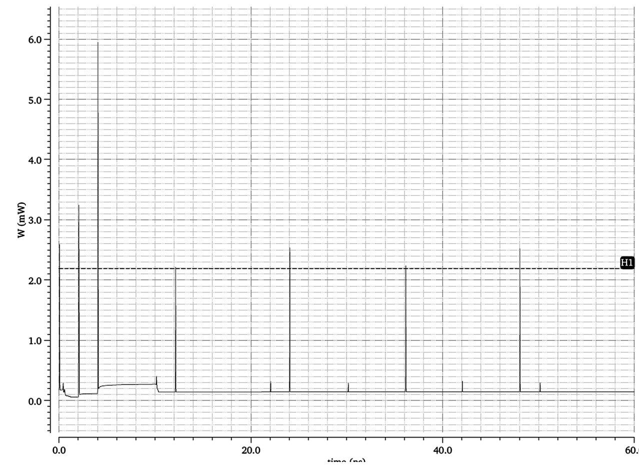

Fig. 8. Power plot for the mismatch case

The Power-Delay Product and Energy-Delay product is computed on a per-bit, per-search basis

$$\text{PDP} = \frac{P_{avg} \times t_{delay}}{N_{bits} \times N_{rows}} = \frac{0.100\,\text{mW} \times 48.8\,\text{ps}}{8 \times 8} = 0.0763 \text{ fJ/bit/search} \tag{1}$$

D. Comparison with Prior Art

Table II presents the performance comparison of the proposed design with the three previously reported precharge-free CAM architectures for equivalent 8×8 NOR-type arrays.

TABLE II
PERFORMANCE COMPARISON OF 8×8 NOR-TYPE MATCHLINE BASED CAM ARCHITECTURES

Architecture	**PF CAM [1]**	**SCPF CAM [3]**	**HSCPF CAM [4]**	**Proposed CAM**
Power (mW)	0.220	0.261	0.213	0.100
Delay (ps)	283.2	101.0	74.0	48.8
PDP (fJ/bit/search)	0.973	0.411	0.246	0.0763

The proposed design achieves the lowest power consumption of 0.100 mW among all compared architectures, representing a 54.54 % reduction over PF CAM [1], a 61.68 % reduction over SCPF CAM [3], and a 53.05 % reduction over the HSCPF CAM architecture [4]. This power improvement is primarily attributed to the suppression of sub-threshold leakage current in the 6T SRAM cross coupled inverters through the selective assignment of HVT transistors, which significantly reduces the static power component that dominates at the 65 nm technology node.

The proposed design also achieves the shortest search delay of 48.8 ps, representing a 82.77 % improvement over PF CAM [1], a 51.68 % improvement over SCPF CAM [3], and a 34.05 % improvement over the HSCPF CAM [4]. This is attributed to the LVT assignment on the search and access transistors, which provides higher drive current and faster match-line discharge upon a mismatch event.

The Power-Delay Product of 0.0763 fJ/bit/search is the best among all compared designs, representing a 92.15 % improvement over PF CAM [1], a 81.43 % improvement over SCPF CAM [3], and a 68.98 % improvement over the HSCPF baseline [4]. These results confirm that transistor-level Dual-VT partitioning, applied selectively based on the functional role of each transistor group within the CAM cell, is an effective strategy for simultaneously reducing leakage power and improving search speed without requiring any changes to the array-level control architecture.

E. Discussion

The power reduction of the proposed design over the HSCPF baseline is attributable primarily to the elimination of sub-threshold leakage in the latch inverters through HVT assignment. Since the four cross coupled inverter transistors contribute the dominant fraction of standby leakage in each 8T cell, replacing them with HVT devices yields a substantial reduction in the static power component.

The search delay improvement is attributed to the LVT search transistors, which provide higher drain current I_{DS} for the same gate overdrive. This allows the match line capacitance to discharge more quickly upon a mismatch event, reducing the detection latency.

The combination of both effects, leakage suppression from HVT inverters and faster discharge from LVT search transistors produces the best PDP result in this class of architecture, confirming that transistor level Dual-VT partitioning is a highly effective and process compatible strategy for CAM optimization at advanced technology nodes.

V. Conclusion

This paper presented a Dual-VT based 8×8 NOR-type CAM array implemented in UMC 65 nm CMOS technology at a supply voltage of 1.2 V, integrating the HSCPF precharge-free architecture with a systematic threshold-voltage assignment strategy. HVT transistors were selectively assigned to the cross-coupled inverters of the 6T SRAM latch to suppress static sub-threshold leakage, while LVT transistors were used for the access and search transistors to maintain fast write and search performance.

Simulations demonstrated an average power of 0.100 mW in the match case and 0.164 mW in the worst-case mismatch scenario, a search delay of 48.8 ps, and a PDP of 0.0763 fJ/bit/search. Compared with the HSCPF baseline, the proposed design achieves 53.05 % power reduction, 34.05 % delay reduction, and 68.98 % PDP reduction.

These results confirm that Dual-VT strategy, when applied selectively based on the functional role of each transistor group within the CAM cell, is an effective and siliconn compatible strategy for simultaneously reducing leakage power and improving search speed without requiring changes to the array-level control architecture. Future work will extend this approach to larger array configurations and ternary CAM implementations at sub-28 nm technology nodes.

Future work will extend this approach to larger array configurations (16×16 and 32×32), investigate ternary CAM (TCAM) implementations with Dual-VT tri-state storage cells, and explore automated multi-VT assignment flows for full-custom CAM synthesis in sub-28 nm technology nodes.

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